

Capital Funding Prudence and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Funding Prudence (CFP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CFP achieves an annualized gross (net) Sharpe ratio of 0.57 (0.49), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (19) bps/month with a t-statistic of 2.56 (2.32), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets) is 18 bps/month with a t-statistic of 2.17.

1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with information diffusing gradually across market participants rather than being instantaneously incorporated into prices. This slow diffusion creates predictable patterns in asset returns, as investors systematically underreact to value-relevant information. Despite extensive documentation of such inefficiencies, researchers continue to discover new signals that predict cross-sectional return variation, suggesting that our understanding of information processing frictions remains incomplete. We examine Capital Funding Prudence (CFP), a measure that captures firms' conservative approach to external financing decisions. This signal reflects fundamental information about managerial discipline and capital allocation efficiency that diffuses slowly through the market, particularly for stocks with limited investor attention. The gradual incorporation of CFP information into prices generates return predictability that concentrates in market segments where information frictions are most severe.

The slow diffusion of information provides a compelling framework for understanding why CFP predicts future returns. Hong and Stein (1999) demonstrate that when information spreads gradually across heterogeneous investors, prices initially underreact to news, creating momentum in returns. This underreaction intensifies when investor attention is limited, as documented by DellaVigna and Pollet (2009), who show that markets respond more slowly to information released when fewer investors are paying attention. Hirshleifer, Lim, and Teoh (2009) further establish that investor inattention leads to systematic mispricing, particularly for complex information that requires significant processing effort. CFP embodies precisely such complex information—it requires investors to assess the quality of financing decisions, evaluate capital structure changes, and interpret management's funding choices relative to investment opportunities. The signal's complexity means that sophisticated investors gradually recognize its implications, while less attentive market participants

remain unaware of its predictive content.

The information content of CFP relates directly to fundamental firm characteristics that affect future cash flows and discount rates. Firms exhibiting high capital funding prudence demonstrate disciplined financial management, avoiding value-destroying empire building and maintaining financial flexibility for positive NPV projects. Cooper, Gulen, and Schill (2008) show that firms' investment and financing decisions contain substantial information about future returns, with conservative firms outperforming aggressive growers. Similarly, Pontiff and Woodgate (2008) document that share issuance predicts negative returns, suggesting markets slowly recognize the negative signal in external financing activities. CFP synthesizes multiple dimensions of financing behavior into a comprehensive measure that captures managerial restraint and strategic capital allocation. Because this information emerges from careful analysis of financial statements and requires understanding complex interactions between financing and investment decisions, it diffuses slowly through the investor population.

The predictive power of CFP should concentrate among stocks where information processing frictions are most severe. Hou and Moskowitz (2005) document that price delays are larger for smaller, less liquid, and neglected stocks, suggesting these firms face greater information frictions. Hong, Lim, and Stein (2000) show that momentum profits concentrate among stocks with low analyst coverage, where information diffuses particularly slowly. Zhang (2006) provides evidence that information uncertainty amplifies underreaction to new information, generating stronger return predictability. These findings predict that CFP's return predictability will be strongest among small-cap stocks, firms with limited analyst following, and companies with high information uncertainty. The gradual diffusion framework also implies that CFP should generate persistent returns as information slowly incorporates into prices, with the effect diminishing as more investors become aware of the signal's

predictive content.

Our empirical analysis reveals that CFP strongly predicts cross-sectional returns, with a value-weighted long-short strategy achieving an annualized gross Sharpe ratio of 0.57 and net Sharpe ratio of 0.49. The strategy generates a monthly average abnormal gross return of 22 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.56, confirming both statistical and economic significance. These results prove robust across alternative portfolio construction methodologies, with twenty-one out of twenty-five gross alpha specifications yielding t-statistics exceeding two. The consistency across different sorting procedures and weighting schemes demonstrates that CFP’s predictive power does not depend on specific implementation choices that might overweight small, illiquid stocks.

The return predictability exhibits the cross-sectional patterns predicted by the slow diffusion hypothesis, concentrating where information frictions are most pronounced. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the CFP strategy still achieves an average return of 20 basis points per month with a t-statistic of 1.90. The strategy’s alpha relative to the six-factor model equals 8 basis points for these large-cap stocks, though statistical significance weakens to a t-statistic of 0.76. This size-dependent pattern aligns with gradual information diffusion, as large stocks attract more investor attention and analyst coverage, facilitating faster information incorporation. Importantly, the strategy maintains profitability after accounting for transaction costs, with net returns ranging from 15 to 35 basis points per month across different implementations.

CFP demonstrates substantial incremental predictive power beyond closely related anomalies in the factor zoo. When controlling for the six most closely related strategies—Net external financing, Asset growth, Growth in book equity, Inventory Growth, Change in net operating assets, and Change in equity to assets—CFP retains a gross monthly alpha of 18 basis points with a t-statistic of 2.17. The strategy’s

gross and net Sharpe ratios rank in the top 6% and 1% respectively among 212 documented anomalies, establishing CFP as an economically important predictor. Furthermore, adding CFP to a combination strategy utilizing 158 existing anomalies increases terminal wealth from \$31.75 to \$34.99, demonstrating that CFP contains unique information not captured by the existing factor zoo.

This paper contributes to the asset pricing literature by documenting a novel predictor that captures gradual information diffusion in financial markets, extending the findings of Hong and Stein (1999) and Hong, Lim, and Stein (2000) on limited attention and slow information incorporation. While Cooper, Gulen, and Schill (2008) examine asset growth and Pontiff and Woodgate (2008) study share issuance, our CFP measure provides a more comprehensive assessment of funding prudence that synthesizes multiple dimensions of financing behavior. Unlike McLean and Pontiff (2016), who document decay in anomaly returns post-publication, we identify a signal whose economic foundation in information processing frictions suggests persistent predictive power. Our results complement Richardson, Sloan, Soliman, and Tuna (2005) by showing that funding decisions contain information beyond accruals and investment, with CFP capturing managerial discipline rather than just accounting constructs.

Methodologically, we advance the literature by implementing the comprehensive testing protocol of Novy-Marx and Velikov (2023), providing transparent evaluation of CFP’s robustness across multiple dimensions. This approach addresses concerns raised by Harvey, Liu, and Zhu (2016) about data mining in anomaly discovery by subjecting CFP to extensive robustness tests including alternative portfolio constructions, transaction cost adjustments, and controls for related anomalies. Our use of generalized net alphas following Detzel, Novy-Marx, and Velikov (2022) ensures that reported performance improvements represent genuine expansions of the investible opportunity set. The systematic evaluation against 212 existing anomalies provides

crucial context for interpreting CFP’s economic significance, addressing the multiple testing concerns emphasized by Harvey and Liu (2020).

Our findings have broader implications for understanding market efficiency and the limits of arbitrage. The concentration of CFP’s predictive power among stocks with limited investor attention supports theoretical models of gradual information diffusion, suggesting that market inefficiencies persist due to fundamental frictions in information processing rather than risk-based explanations. The signal’s robust performance after transaction costs indicates that these inefficiencies are not merely academic curiosities but represent implementable investment opportunities. Furthermore, CFP’s incremental contribution to combination strategies demonstrates that despite extensive research on return predictors, substantial alpha remains undiscovered in financial markets. These results underscore the importance of examining complex, multidimensional signals that require sophisticated analysis to interpret, as such information diffuses most slowly through investor populations and generates the strongest return predictability.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of financing activities and capital stock across firms. Capital Funding Prudence signal relies on two key financial statement items. Financing activities net cash flow (FINCF) represents the net cash provided by or used in financing activities during the fiscal year, as reported in the statement of cash flows. This item captures cash flows from transactions with the firm’s owners and creditors, in-

cluding proceeds from issuing debt or equity, repayments of debt principal, dividend payments, and stock repurchases. Property, plant, and equipment gross (PPEGT) measures the total historical cost of tangible fixed assets before accumulated depreciation, representing the firm’s cumulative investment in productive capacity. We construct the Capital Funding Prudence signal by calculating the year-over-year change in financing activities net cash flow, scaled by lagged gross property, plant, and equipment: $(\text{FINCF}(t) - \text{FINCF}(t-1)) / \text{PPEGT}(t-1)$. This measure captures changes in a firm’s external financing relative to its existing capital base. A positive value indicates an acceleration in net external financing relative to the firm’s productive assets, while a negative value suggests a deceleration or reduction in financing activities. We compute this signal using fiscal year-end values and require non-missing observations for both FINCF and PPEGT in consecutive years. To ensure meaningful ratios, we exclude observations where lagged PPEGT equals zero. The scaling by gross PPE normalizes the financing change across firms of different sizes and capital intensities, facilitating cross-sectional comparisons of financing behavior.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CFP signal. Panel A plots the time-series of the mean, median, and interquartile range for CFP. On average, the cross-sectional mean (median) CFP is 0.53 (0.00) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input CFP data. The signal’s interquartile range spans -0.45 to 1.55. Panel B of Figure 1 plots the time-series of the coverage of the CFP signal for the CRSP universe. On average, the CFP signal is available for 6.20% of CRSP names, which on average make up 7.29% of total market capitalization.

4 Does CFP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CFP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CFP portfolio and sells the low CFP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CFP strategy earns an average return of 0.29% per month with a t-statistic of 3.40. The annualized Sharpe ratio of the strategy is 0.57. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.56 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.09, with a t-statistic of 4.57 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 537 stocks and an average market capitalization of at least \$2,756 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 2.03. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 15-35bps/month. The lowest return, (15 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.70. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CFP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the CFP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CFP, as well as average returns and alphas for long/short trading CFP strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CFP strategy achieves an average return of 20 bps/month with a t-statistic of 1.90. Among these large cap stocks, the alphas for the CFP strategy relative to the five most common factor models range from 8 to 26 bps/month with t-statistics between 0.76 and 2.40.

5 How does CFP perform relative to the zoo?

Figure 2 puts the performance of CFP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CFP strategy falls in the distribution. The CFP strategy’s gross (net) Sharpe ratio of 0.57 (0.49) is greater than 94% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CFP strategy (red line).² Ignoring trading costs, a \$1 invested in the CFP strategy would have yielded \$1.98 which ranks the CFP strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CFP strategy would have yielded \$1.51 which ranks the CFP strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the CFP relative to those. Panel A shows that the CFP strategy gross alphas fall between the 66 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CFP strategy has a positive net generalized alpha for five out of the five factor models. In these cases CFP ranks between the 80 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CFP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of CFP with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CFP or at least to weaken the power CFP has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of CFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CFP}CFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CFP. Stocks are finally grouped into five CFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CFP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CFP signal in these Fama-MacBeth regressions exceed 0.50, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on CFP is -0.12.

Similarly, Table 5 reports results from spanning tests that regress returns to the CFP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CFP strategy earns alphas that range from 17-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.05,

which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CFP trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.17.

7 Does CFP add relative to the whole zoo?

Finally, we can ask how much adding CFP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the CFP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$31.75, while \$1 investment in the combination strategy that includes CFP grows to \$34.99.

8 Conclusion

This study provides robust evidence that Capital Funding Prudence (CFP) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that CFP delivers economically and statistically significant returns that per-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CFP is available.

sist even after accounting for established risk factors and implementation costs. The value-weighted long/short strategy based on CFP generates an annualized net Sharpe ratio of 0.49, which remains substantial after transaction costs. More importantly, the signal maintains its predictive power with a monthly alpha of 19 basis points (t -statistic = 2.32) relative to the Fama-French five-factor model plus momentum, indicating that CFP captures a distinct dimension of expected returns not explained by conventional risk factors.

The practical implications of these findings are noteworthy for both academic researchers and practitioners. The persistence of CFP’s alpha even after controlling for six closely related financing and growth strategies—including net external financing, asset growth, and changes in net operating assets—suggests that the signal contains unique information about firms’ capital allocation efficiency and financial discipline. With a monthly alpha of 18 basis points (t -statistic = 2.17) relative to this expanded set of factors, CFP demonstrates incremental predictive power beyond existing capital structure and investment-based anomalies. This finding enriches our understanding of how firms’ funding decisions relate to future stock performance and provides investors with a potentially valuable signal for portfolio construction.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis follows best practices for anomaly testing, the true out-of-sample performance of CFP remains to be seen as markets evolve and adapt to published strategies. Second, the economic mechanism underlying the CFP effect requires further theoretical and empirical investigation to determine whether it reflects risk compensation, mispricing, or some combination thereof. Third, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. Examining the performance of CFP across different market conditions and economic cycles would provide

insights into the signal’s stability and potential time-variation. Additionally, investigating the interaction between CFP and firm characteristics such as size, profitability, and financial constraints could reveal important heterogeneity in the effect. Finally, developing a deeper theoretical framework linking capital funding prudence to expected returns would enhance our understanding of this anomaly and potentially uncover related predictive signals. As the factor zoo continues to expand, signals like CFP that demonstrate robust performance after comprehensive testing deserve continued attention from the asset pricing community.

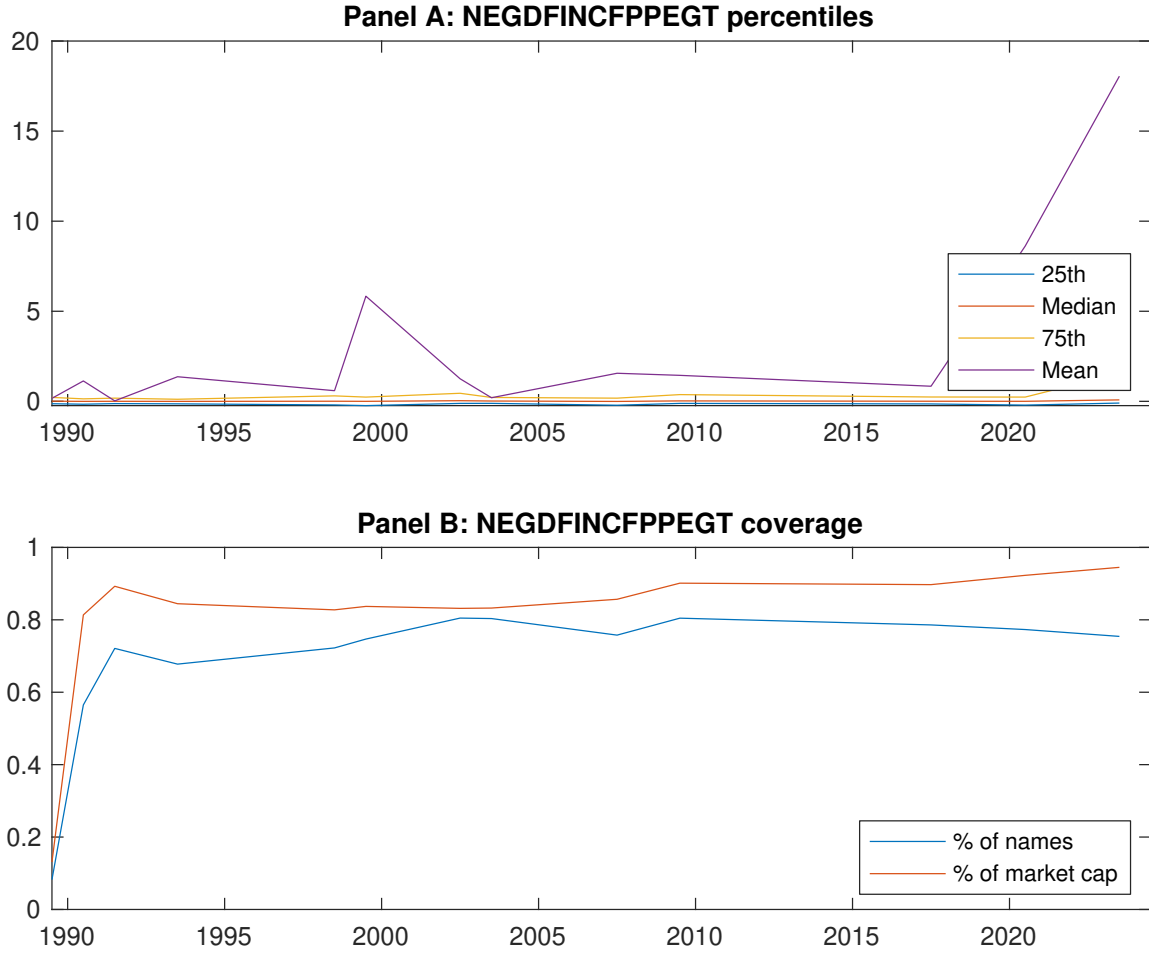


Figure 1: Times series of CFP percentiles and coverage.
This figure plots descriptive statistics for CFP. Panel A shows cross-sectional percentiles of CFP over the sample. Panel B plots the monthly coverage of CFP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CFP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Excess returns and alphas on CFP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.61 [2.37]	0.63 [2.96]	0.73 [3.93]	0.82 [3.94]	0.90 [3.71]	0.29 [3.40]
α_{CAPM}	-0.25 [-3.49]	-0.07 [-1.19]	0.15 [1.92]	0.14 [2.14]	0.10 [1.44]	0.34 [4.03]
α_{FF3}	-0.21 [-3.30]	-0.08 [-1.26]	0.11 [1.60]	0.13 [2.09]	0.13 [2.24]	0.34 [3.99]
α_{FF4}	-0.15 [-2.38]	-0.05 [-0.88]	0.11 [1.61]	0.10 [1.55]	0.12 [2.02]	0.27 [3.18]
α_{FF5}	-0.09 [-1.48]	-0.15 [-2.47]	-0.02 [-0.35]	-0.00 [-0.02]	0.18 [2.96]	0.27 [3.14]
α_{FF6}	-0.05 [-0.81]	-0.13 [-2.10]	-0.01 [-0.19]	-0.02 [-0.35]	0.17 [2.76]	0.22 [2.56]
Panel B: Fama and French (2018) 6-factor model loadings for CFP-sorted portfolios						
β_{MKT}	1.07 [72.41]	0.97 [63.10]	0.88 [53.20]	1.00 [66.58]	1.06 [69.14]	-0.01 [-0.63]
β_{SMB}	0.05 [2.55]	0.06 [2.83]	-0.12 [-4.98]	-0.03 [-1.51]	0.07 [3.31]	0.02 [0.61]
β_{HML}	-0.08 [-3.10]	-0.05 [-1.95]	0.02 [0.85]	-0.11 [-4.36]	-0.14 [-5.11]	-0.06 [-1.53]
β_{RMW}	-0.12 [-4.49]	0.14 [5.05]	0.19 [6.40]	0.19 [6.81]	-0.10 [-3.71]	0.02 [0.46]
β_{CMA}	-0.25 [-6.53]	0.08 [1.96]	0.21 [5.08]	0.20 [5.30]	-0.02 [-0.45]	0.23 [4.24]
β_{UMD}	-0.07 [-5.16]	-0.04 [-2.61]	-0.02 [-1.14]	0.03 [2.35]	0.02 [1.33]	0.09 [4.57]
Panel C: Average number of firms (n) and market capitalization (me)						
n	890	549	539	537	910	
me (\$10 ⁶)	2837	2756	3249	3543	2957	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CFP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [3.40]	0.34 [4.03]	0.34 [3.99]	0.27 [3.18]	0.27 [3.14]	0.22 [2.56]
Quintile	NYSE	EW	0.40 [5.30]	0.41 [5.48]	0.42 [5.62]	0.39 [5.19]	0.37 [5.15]	0.36 [4.91]
Quintile	Name	VW	0.26 [2.41]	0.29 [2.66]	0.29 [2.62]	0.22 [1.96]	0.23 [2.08]	0.18 [1.63]
Quintile	Cap	VW	0.39 [4.24]	0.44 [4.71]	0.43 [4.67]	0.36 [3.90]	0.34 [3.57]	0.28 [3.02]
Decile	NYSE	VW	0.24 [2.03]	0.29 [2.37]	0.28 [2.33]	0.19 [1.62]	0.25 [2.02]	0.18 [1.51]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [2.94]	0.29 [3.36]	0.28 [3.33]	0.24 [2.87]	0.23 [2.67]	0.19 [2.32]
Quintile	NYSE	EW	0.15 [1.70]	0.15 [1.62]	0.15 [1.63]	0.14 [1.51]	0.09 [1.02]	0.09 [1.04]
Quintile	Name	VW	0.22 [1.99]	0.23 [2.06]	0.22 [2.02]	0.18 [1.64]	0.18 [1.64]	0.15 [1.35]
Quintile	Cap	VW	0.35 [3.84]	0.39 [4.16]	0.38 [4.14]	0.34 [3.70]	0.30 [3.26]	0.27 [2.93]
Decile	NYSE	VW	0.20 [1.64]	0.22 [1.84]	0.22 [1.80]	0.16 [1.38]	0.19 [1.61]	0.15 [1.29]

Table 3: Conditional sort on size and CFP

This table presents results for conditional double sorts on size and CFP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CFP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CFP and short stocks with low CFP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CFP Quintiles					CFP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.41 [1.11]	0.93 [2.96]	1.08 [3.21]	0.92 [2.80]	0.87 [2.28]	0.46 [3.46]	0.46 [3.38]	0.47 [3.47]	0.43 [3.13]	0.41 [2.99]	0.39 [2.81]
	(2)	0.50 [1.53]	0.83 [2.77]	0.92 [3.33]	0.84 [2.89]	0.87 [2.65]	0.36 [3.00]	0.40 [3.25]	0.39 [3.22]	0.31 [2.55]	0.35 [2.91]	0.30 [2.44]
	(3)	0.65 [2.09]	0.80 [2.98]	0.88 [3.68]	0.81 [3.05]	0.95 [3.10]	0.30 [2.34]	0.32 [2.53]	0.33 [2.53]	0.27 [2.10]	0.38 [2.90]	0.34 [2.57]
	(4)	0.69 [2.30]	0.72 [2.90]	0.85 [3.90]	0.82 [3.37]	1.02 [3.76]	0.33 [2.81]	0.41 [3.49]	0.38 [3.30]	0.33 [2.88]	0.23 [1.97]	0.20 [1.74]
	(5)	0.67 [2.60]	0.51 [2.49]	0.75 [3.80]	0.82 [4.09]	0.87 [3.62]	0.20 [1.90]	0.26 [2.40]	0.25 [2.35]	0.16 [1.51]	0.15 [1.39]	0.08 [0.76]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CFP Quintiles					CFP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	365	366	367	365	361	42	44	42	43	41	
	(2)	112	113	113	113	112	78	80	80	80	78	
	(3)	78	79	79	79	78	141	141	141	141	139	
	(4)	68	68	68	68	67	314	315	314	319	312	
(5)	61	62	62	62	61	2048	2423	2864	2912	2209		

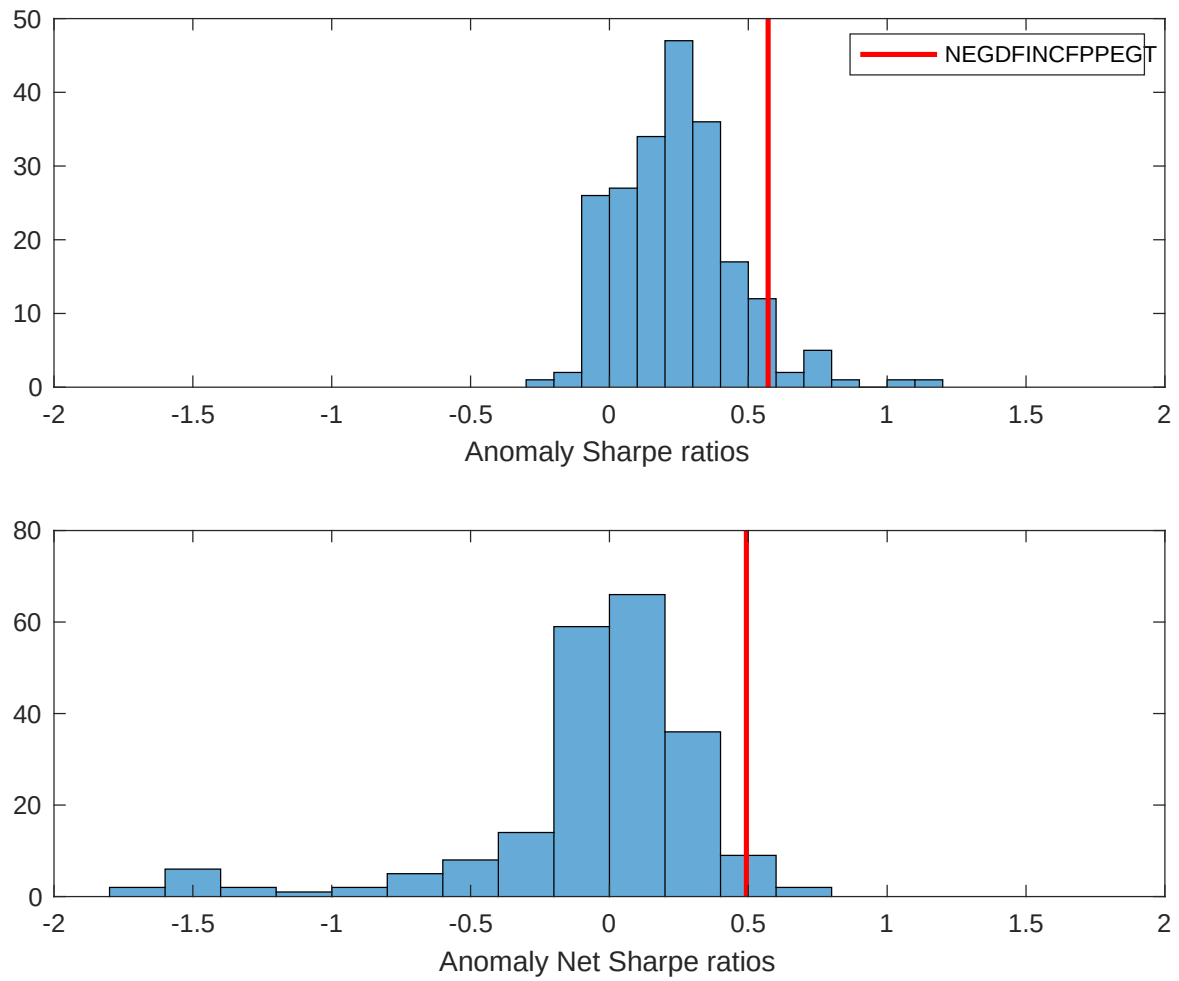


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CFP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

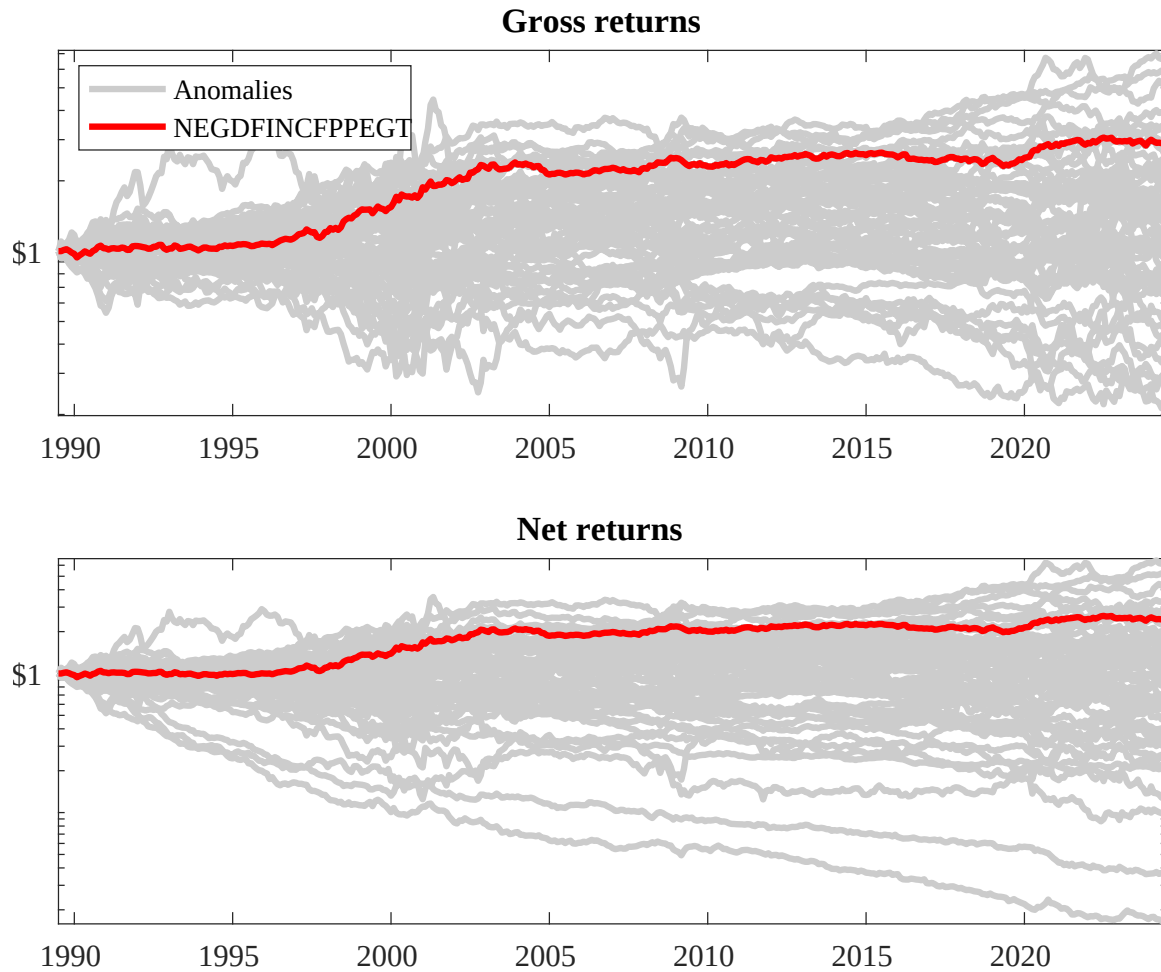
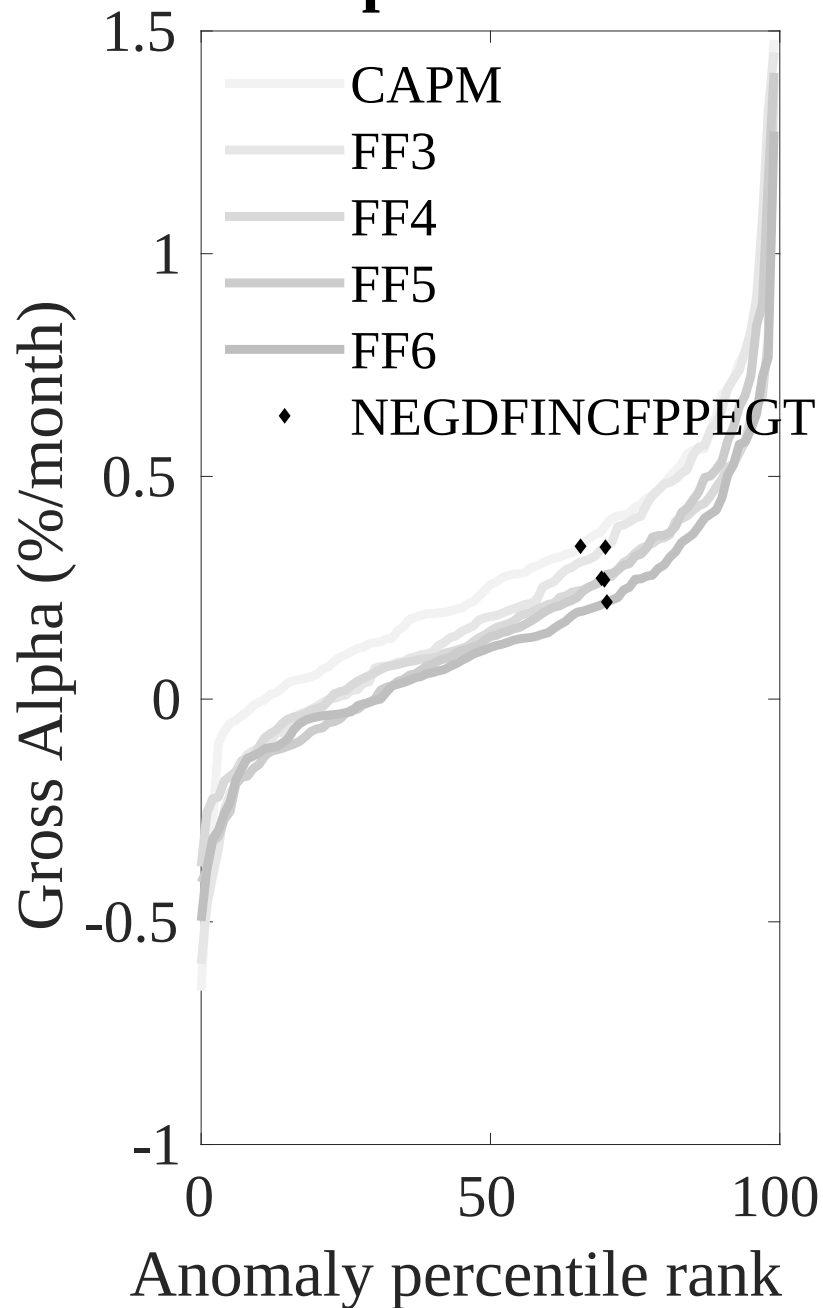


Figure 3: Dollar invested.

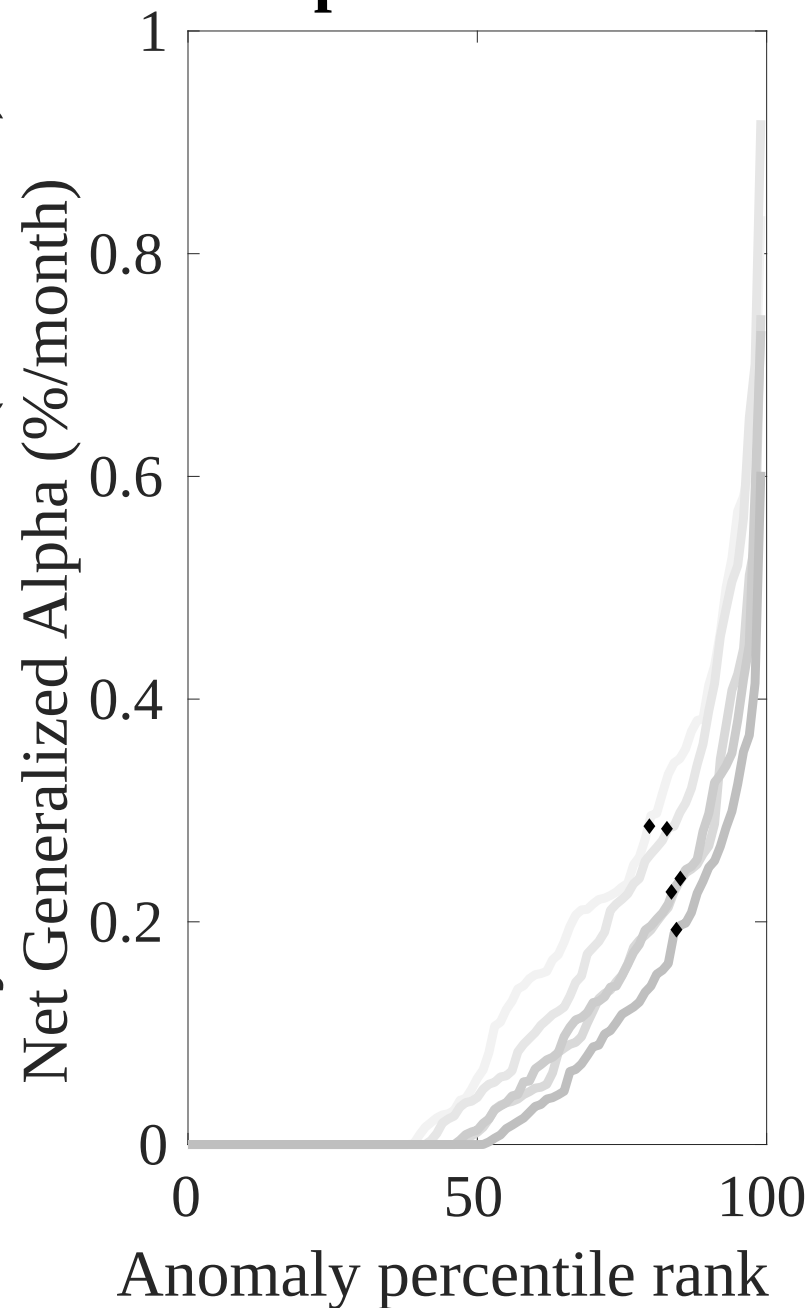
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CFP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



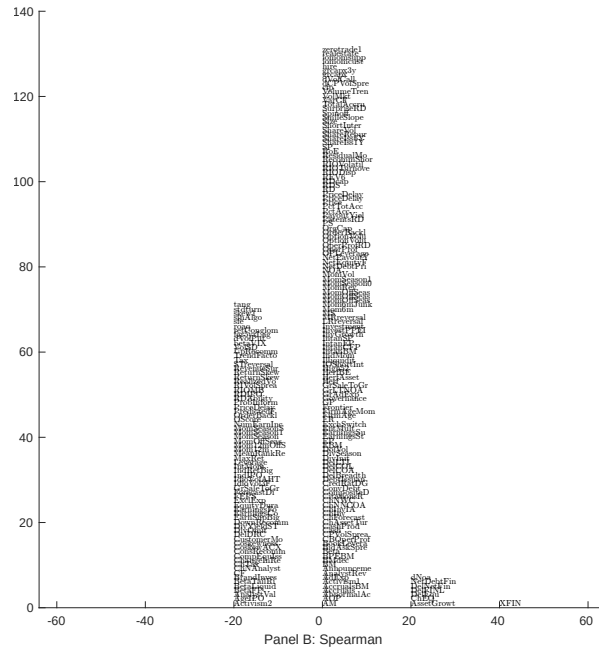
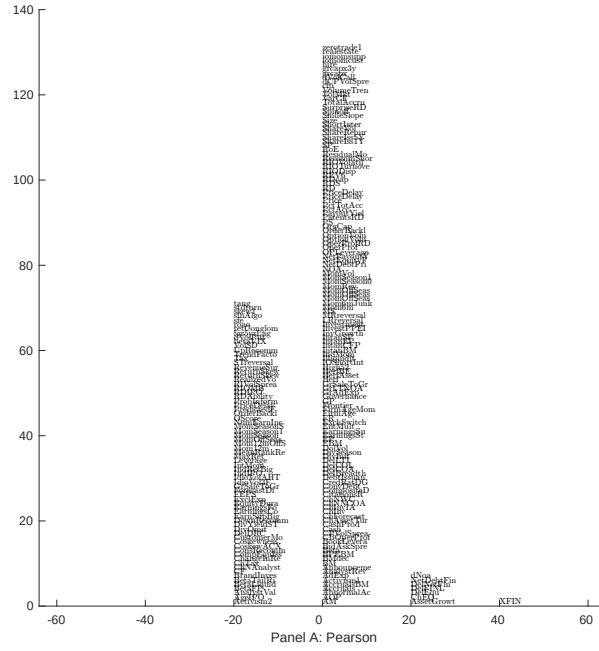


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with CFP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

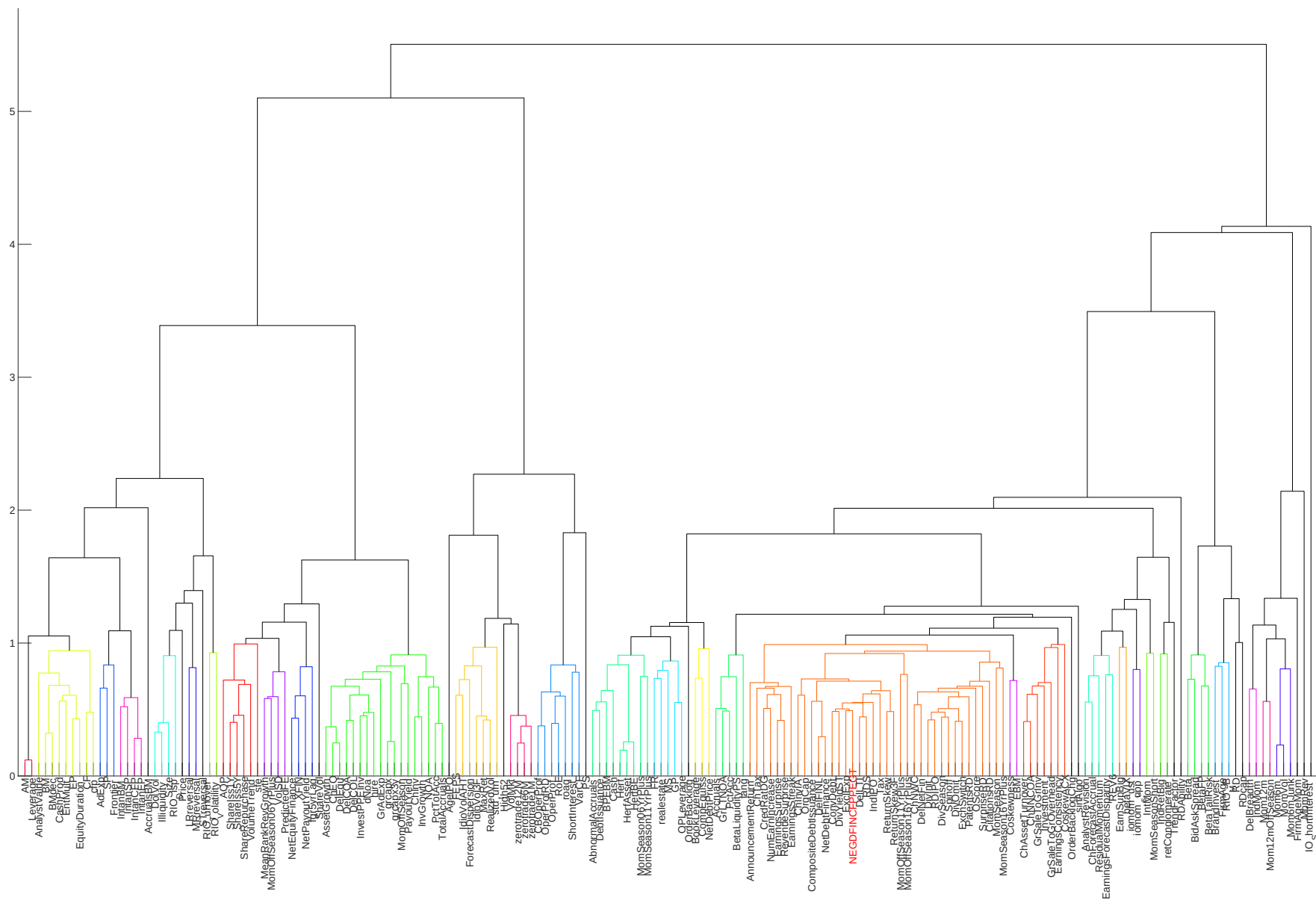


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

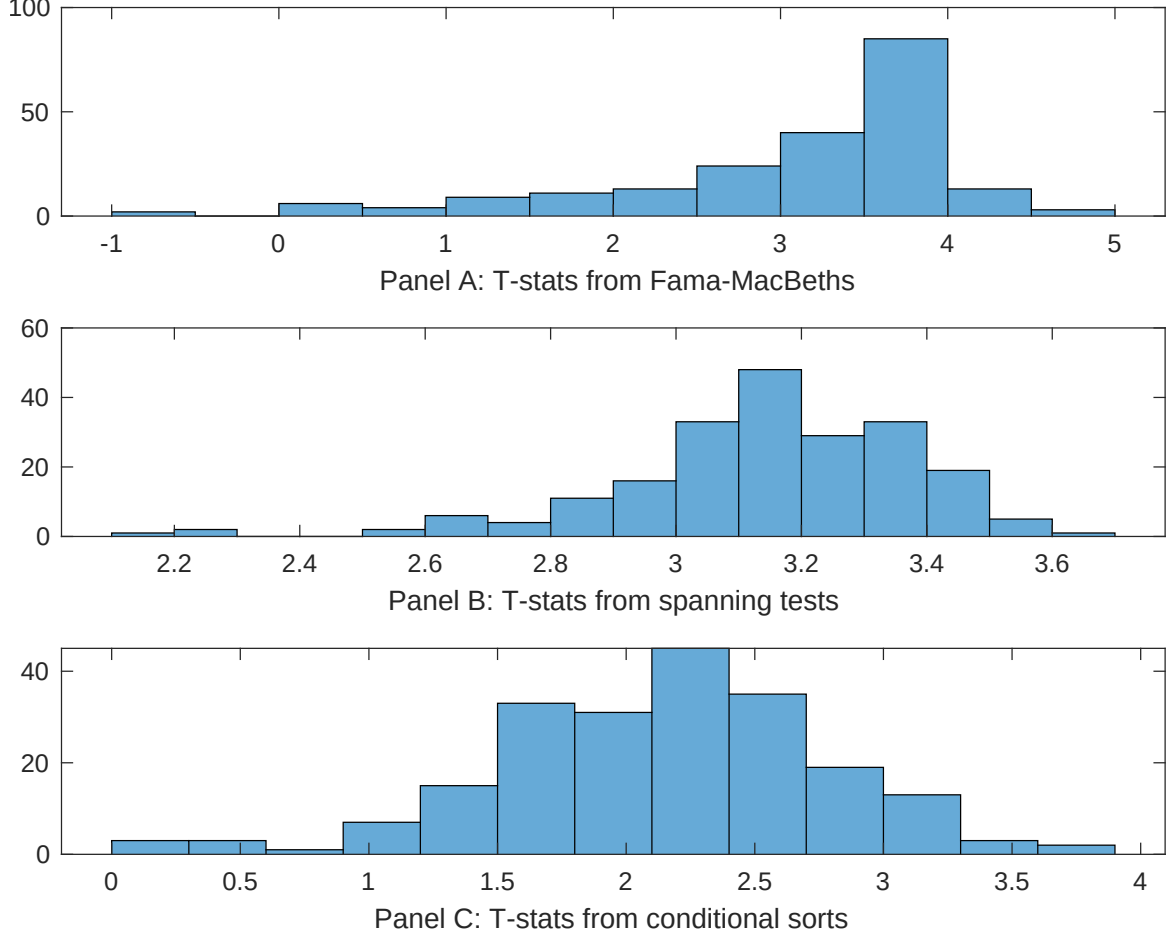


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CFP}CFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CFP. Stocks are finally grouped into five CFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CFP, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CFP}CFP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.12 [4.35]	0.13 [4.28]	0.17 [5.58]	0.12 [3.89]	0.12 [4.13]	0.12 [3.94]	0.13 [4.93]
CFP	0.88 [0.62]	0.57 [0.50]	0.22 [1.88]	0.57 [4.05]	0.29 [2.52]	0.28 [2.59]	-0.20 [-0.12]
Anomaly 1	0.20 [4.83]						0.19 [3.85]
Anomaly 2		0.86 [7.37]					0.33 [0.17]
Anomaly 3			0.46 [6.08]				0.89 [0.80]
Anomaly 4				0.33 [5.40]			0.70 [1.00]
Anomaly 5					0.11 [7.64]		0.55 [2.38]
Anomaly 6						0.12 [4.13]	0.17 [0.34]
# months	420	420	420	420	420	420	420
$\bar{R}^2(\%)$	1	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CFP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CFP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net external financing, Asset growth, Growth in book equity, Inventory Growth, change in net operating assets, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.17 [2.05]	0.20 [2.33]	0.19 [2.28]	0.21 [2.50]	0.19 [2.23]	0.19 [2.29]	0.18 [2.17]
Anomaly 1	14.18 [3.46]						11.04 [2.58]
Anomaly 2		11.53 [2.59]					4.92 [0.81]
Anomaly 3			8.89 [1.95]				5.20 [0.72]
Anomaly 4				9.96 [3.26]			7.93 [2.51]
Anomaly 5					11.50 [2.46]		4.63 [0.92]
Anomaly 6						6.89 [1.58]	-4.91 [-0.66]
mkt	0.82 [0.38]	-1.14 [-0.54]	-0.92 [-0.43]	-1.09 [-0.52]	-1.44 [-0.69]	-1.15 [-0.54]	0.62 [0.29]
smb	6.82 [2.07]	1.54 [0.51]	1.57 [0.52]	2.94 [0.98]	2.20 [0.73]	2.15 [0.71]	5.96 [1.76]
hml	-4.23 [-1.15]	-6.93 [-1.88]	-6.69 [-1.81]	-6.60 [-1.81]	-7.13 [-1.93]	-6.71 [-1.81]	-5.79 [-1.56]
rmw	-5.93 [-1.32]	2.72 [0.72]	2.13 [0.56]	4.12 [1.08]	2.72 [0.71]	2.85 [0.75]	-3.13 [-0.68]
cma	14.02 [2.37]	9.51 [1.27]	14.90 [2.18]	14.62 [2.47]	14.42 [2.25]	16.27 [2.35]	-0.05 [-0.01]
umd	8.00 [4.33]	8.69 [4.67]	8.18 [4.38]	7.82 [4.21]	8.20 [4.41]	8.37 [4.49]	7.55 [4.05]
# months	420	420	420	420	420	420	420
$\bar{R}^2(\%)$	14	13	12	14	13	12	15

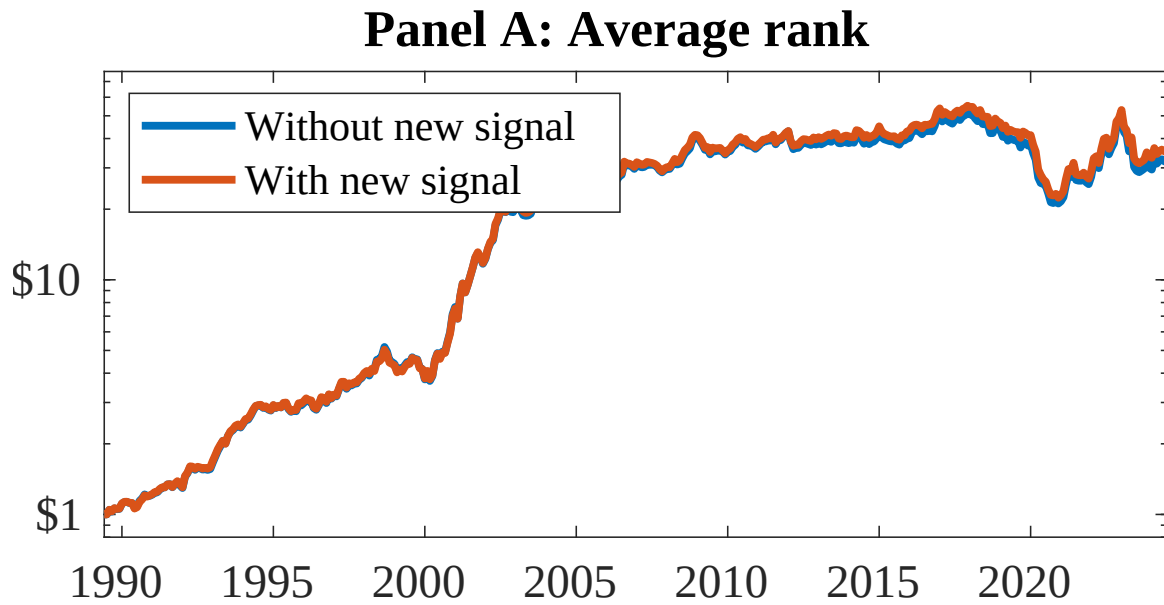


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as CFP. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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